From Paid APIs to Local GPUs: Zero-Shot Consumer Financial Complaint Classification with Open-Source LLMs

Muhammet Tan[1] and Halit Bakır[1*]

[1*]Department of Computer Engineering, Sivas University of Science and Technology, Sivas, 58010, Turkey.

*Corresponding author(s). E-mail(s): halit.bakir@sivas.edu.tr;
Contributing authors: muhammettan28@gmail.com;

Abstract

Automated classification of consumer financial complaints has significant implications for regulatory oversight and customer service prioritisation. While recent studies have demonstrated strong zero-shot performance using large proprietary language model (LLM) APIs, those approaches incur substantial and recurring monetary costs. In this paper we investigate whether open-source, locally deployed LLMs can match the classification quality of their paid-API counterparts at zero API cost. We benchmark eleven open-weight models (six standard and five reasoning) on the Consumer Financial Protection Bureau (CFPB) five-category zero-shot task, across two experimental conditions (balanced and imbalanced), three random seeds, and two narrative length limits (250 and 500 characters). Our best model, `gemma4:12b`, achieves a mean weighted F1 of $\mathbf{0.774 \pm 0.016}$ on the balanced evaluation — matching paid GPT-4.5 (F1 0.7763) and approaching o1 (F1 0.7764) at zero cost. Notably, this top result comes from a 12B standard (non-reasoning) model, challenging the view that reasoning models are necessary for structured classification tasks. We analyse off-label prediction behaviour and provide deployment recommendations based on the latency–accuracy trade-off observed across all eleven models.

Keywords: consumer complaint classification, large language models, zero-shot learning, open-source LLMs, CFPB, reasoning models, natural language processing

Manuscript received June 2026.

1 Introduction

The Consumer Financial Protection Bureau (CFPB) publishes a public database of consumer complaints submitted against financial institutions in the United States. The dataset spans dozens of product categories and hundreds of thousands of narratives, making it one of the most accessible and practically relevant corpora for natural language processing (NLP) research in the financial domain. Accurately routing and categorising these complaints has direct implications for regulatory efficiency, bank compliance teams, and consumer advocacy organisations: a miscategorised complaint may be forwarded to the wrong department, delayed, or deprioritised [1–3].

Unlike many benchmark text-classification datasets, consumer complaint narratives are not merely short topic labels embedded in clean prose. They often contain informal descriptions of stressful financial events, institution names, dates, partial account details, and overlapping product issues. A single complaint may mention credit reporting, debt collection, and a credit card dispute in the same narrative, while the supervised label assigns only one product category. This ambiguity makes the task more realistic than many standard NLP benchmarks and also more valuable as a test bed for instruction-following models: a useful system must map noisy, human-written narratives to a constrained regulatory taxonomy without inventing new labels.

The practical motivation is equally strong. Complaint triage is a high-volume activity in which automation can reduce manual screening time, support earlier detection of recurring institutional problems, and provide consistent first-pass routing [4]. At the same time, financial complaint text is sensitive. Even when datasets are public, a deployable classification workflow for banks, regulators, or research laboratories may need to operate without sending narratives to third-party services. This makes local inference, auditability, and reproducible infrastructure more than implementation details; they are part of the methodological problem.

Traditional approaches to complaint classification relied on bag-of-words representations, support vector machines, and logistic regression. The emergence of BERT [5] and its domain-adapted successors marked a step change, enabling much richer semantic representations and pushing classification accuracy above 90 % in fine-tuned settings [6]. More recently, the advent of instruction-following LLMs such as GPT-3 [7] and its successors opened the possibility of *zero-shot* classification: a model is simply given the candidate categories in a prompt and is asked to assign the most appropriate one, requiring no task-specific training data at all [8, 9].

For consumer complaint analysis, this zero-shot setting is attractive because financial taxonomies evolve and labelled data may be costly to curate for every new jurisdiction, product definition, or institutional workflow. However, zero-shot classification also shifts the main engineering burden from model training to prompt adherence, output parsing, and deployment cost. A model that returns a plausible but unsupported category such as `credit_collection` may look linguistically competent while still failing the regulatory routing task. Therefore, evaluating off-label behaviour is essential alongside accuracy and F1.

Roumeliotis et al. [10] conducted a comprehensive zero-shot benchmark of 14 state-of-the-art proprietary and commercial models — including GPT-4o, GPT-4.5, Claude 3.5/3.7, Gemini 2.0, o1, o3-mini, and DeepSeek (API) — on the same CFPB

dataset and five-category task. They reported that the best model, OpenAI's o1, achieved an accuracy of 0.777 and an F1 of 0.7764, at a cost of $24.24 per thousand predictions. Their study provided strong evidence that reasoning-enabled LLMs can outperform standard chat models on this structured classification task. However, a critical limitation of their work is that *every* model was accessed through a paid commercial API, creating a barrier to reproduction and deployment in low-budget or data-sensitive environments.

This leaves an important open question: can open-weight models, served locally on shared research infrastructure, deliver competitive zero-shot complaint classification without API access? The answer is not obvious. Smaller local models are cheaper to run but may be weaker at following strict JSON-style instructions. Reasoning models may improve disambiguation and output discipline, but they also generate longer traces and increase latency. In addition, prior balanced benchmarks do not show whether the same ranking survives under the natural class imbalance of the CFPB corpus, where a majority `credit_reporting` class can inflate aggregate metrics.

The present paper addresses this gap directly. Our contributions are:

1. We benchmark eleven open-source, open-weight LLMs — `gemma2:9b`, `gemma4:12b`, `mistral:7b`, `mistral-nemo:12b`, `llama3.1:8b`, `phi4:14b`, `deepseek-r1:8b`, `deepseek-r1:14b`, `qwen3:8b`, `qwen3:14b`, and `gpt-oss:20b` — on the same five-category CFPB zero-shot task, at **zero API cost**. This expanded model set includes matched 12–14B variants for each architecture family, enabling a fair cross-scale comparison.
2. We document a complete, reproducible local inference pipeline using Ollama on an Ubuntu workstation, including the hardware and software parameters needed to replicate our results.
3. We conduct a second experiment using the natural (imbalanced) class distribution, evaluating whether performance patterns observed under artificial balance hold in a more realistic scenario.
4. We provide a direct quantitative comparison with the paid-API results of Roumeliotis et al. [10], showing that our best open-source model exceeds the reported GPT-4o score under the same evaluation protocol and remains competitive with more expensive API-based alternatives.
5. We analyse off-label prediction behaviour — instances where models invent labels outside the five prescribed categories — as a proxy for instruction-following reliability.

The remainder of this paper is structured as follows. Section 2 surveys the related literature. Section 3 describes the dataset and preprocessing pipeline. Section 4 details the experimental setup. Section 5 presents results for both experiments. Section 6 discusses the findings. Section 7 concludes with directions for future work.

2 Related Work

2.1 From Rule-Based Systems to Transformer Models

The challenge of automatically classifying consumer complaints has a rich history that mirrors the broader trajectory of NLP research. Early systems relied on keyword matching and hand-crafted feature engineering paired with classical classifiers such as support vector machines and naïve Bayes. While effective for narrow, well-defined complaint taxonomies, these approaches proved brittle in the face of the linguistic diversity found in real-world financial complaint corpora [3, 11]. Lilleberg et al. [12] demonstrated that pairing support vector machines with Word2Vec embeddings improves text classification over purely bag-of-words features, yet even these enriched representations required domain-specific engineering and large labelled corpora.

The introduction of BERT [5] triggered a significant paradigm shift. Contextual word embeddings, obtained by pre-training on large general-domain corpora and then fine-tuning on downstream tasks, dramatically reduced the gap between human and machine performance on text classification. Raju et al. [11] applied BERTopic [13], FinBERT, DistilBERT, and RoBERTa to a corpus of 684,653 CFPB complaints, demonstrating that transformer-based topic models could discover latent complaint themes without manual annotation. Vairetti et al. [6] used the Spanish BERT variant BETO for complaint prioritisation, achieving 92.1 % accuracy, illustrating that the BERT paradigm transfers well across languages and regulatory contexts. Das et al. [1] pushed further by coupling transformer-based classifiers with explainability techniques, enabling practitioners to understand *why* a complaint was assigned a given category — a requirement for regulatory compliance. Liu et al. [2] proposed a structured LLM-based categorisation framework (CCF) with real-time monitoring, demonstrating that modern LLMs can serve as end-to-end complaint classification pipelines with minimal domain customisation.

2.2 The LLM Era and the Zero-Shot Paradigm

The release of GPT-3 [7] introduced the few-shot and zero-shot paradigms, in which a large pre-trained language model can perform novel tasks from a natural language description alone. This capability fundamentally changed the cost calculus of NLP system development: instead of collecting thousands of labelled examples and retraining a model, practitioners could now describe a task in a prompt and obtain reasonable predictions immediately. Zhao et al. [14] provide a comprehensive survey of the capabilities and limitations of modern LLMs, noting that instruction-following performance scales strongly with model size and training quality.

In the financial complaint domain, Correa et al. [15] demonstrated that generative AI can simultaneously classify, summarise, and draft responses to CFPB complaints, achieving 88 % classification accuracy in a zero-shot setting. This result was notable because it required no fine-tuning whatsoever, confirming that the rich language representations encoded in large-scale pre-training are sufficient for this task. Extending the zero-shot paradigm to service industries, Roumeliotis et al. [16] compared GPT-4 Omni and BERT for customer review classification in tourism, finding that GPT-4o

achieved 67 % accuracy in a zero-shot setting versus 60.6 % for fine-tuned BERT — an early indication that zero-shot LLMs can challenge fine-tuned specialised models. In a related e-commerce study, Roumeliotis et al. [17] compared GPT-3.5 and LLaMA-2 alongside BERT and RoBERTa for product review sentiment classification, confirming that LLMs achieve competitive performance across diverse consumer feedback domains both before and after fine-tuning. Li et al. [18] further showed, through a comprehensive customer service benchmark (TeleEval-CS), that fine-tuned open-source LLMs can match proprietary counterparts when evaluated on domain-specific dialogue tasks, motivating our investigation of open-weight models.

A practically important contextual finding is that the CFPB complaint corpus itself is changing. Shin et al. [19] analysed over 1.1 million CFPB complaints and found that approximately 9.8 % of complaints submitted after the public release of ChatGPT exhibit strong evidence of LLM assistance. Liang et al. [20] independently estimated that roughly 18 % of recent complaints are LLM-assisted. These findings underscore the increasing entanglement of generative AI and financial consumer protection, making robust automated classification even more important.

2.3 Reasoning Models and the State of the Art

The most recent development in this space has been the introduction of chain-of-thought reasoning models [21], which generate intermediate reasoning steps before arriving at a final answer. DeepSeek-AI [22] demonstrated that open-source models trained with careful scaling laws can match GPT-3.5 tier performance, establishing the viability of open-weight alternatives before the reasoning model era. Building on this foundation, Guo et al. [23] introduced DeepSeek-R1, an open-weight model trained via reinforcement learning to produce structured reasoning traces, achieving performance competitive with proprietary frontier models on a range of benchmarks. Similarly, the Qwen3 series [24] introduced an explicit thinking mode that can be toggled per query.

Roumeliotis et al. [10] brought this reasoning paradigm directly to the complaint classification setting. Testing 14 models on a balanced sample of 1,000 CFPB complaints across five categories, they found that o1 (0.777 accuracy, F1 0.7764) and GPT-4.5 (0.774, F1 0.7763) outperformed their non-reasoning counterparts, and that reasoning models as a class exhibited fewer off-label predictions. Their study is the closest antecedent to the present work and provides the benchmark against which our results are measured.

Concurrently, Bulgakov and Turchin [25] explored DeepSeek-R1-Distill-LLaMA-8B for binary prediction of monetary relief outcomes from CFPB complaints in a zero-shot setting, demonstrating that distilled reasoning models retain competitive performance while fitting on a single consumer-grade GPU. Güllü et al. [26] fine-tuned LLaMA-3.1-8B on the same five-category CFPB task used in our study, reaching 89.99 % accuracy and establishing an approximate fine-tuning performance ceiling for that model family.

2.4 The Open-Source Gap

Despite these advances, a critical gap remains in the literature. Every paper benchmarking LLMs on CFPB complaint classification has relied either on paid commercial

APIs or on fine-tuned models that require substantial labelled training data. Neither approach is accessible to practitioners operating under strict data-governance constraints, limited budgets, or air-gapped computing environments. The present paper fills this gap by restricting itself entirely to open-weight models deployed locally on commodity workstation hardware, thereby demonstrating that competitive zero-shot complaint classification is achievable without any API cost or proprietary model access.

3 Dataset and Preprocessing

3.1 Data Source

We use the CFPB Consumer Complaints dataset originally published by Tiwari [27] on Kaggle, which is a processed subset of the public CFPB Consumer Complaint Database. The dataset covers complaints submitted between March 2020 and March 2021 and contains three columns: an unnamed index, `product` (the category label), and `narrative` (the free-text complaint). The raw file comprises 162,421 records consolidated to five target product categories: `retail_banking`, `credit_reporting`, `credit_card`, `mortgages_and_loans`, and `debt_collection`. This is the same dataset and label set used by Roumeliotis et al. [10], ensuring direct comparability.

3.2 Class Distribution

The original data are heavily imbalanced. After removing records with empty `product` or `narrative` fields (10 rows removed, 162,411 remaining) and discarding narratives exceeding 500 characters (resulting in 103,915 usable records), the class distribution shifts to: `credit_reporting` 63.6 %, `debt_collection` 14.1 %, `mortgages_and_loans` 7.7 %, `credit_card` 7.5 %, and `retail_banking` 7.1 %. Before the length filter, the same five-category corpus contains 162,411 records with `credit_reporting` accounting for 56.1 % of the data. Fig. 1 illustrates how this skew propagates into the experimental subsets: the imbalanced condition retains the dominant `credit_reporting` class at 51.1 %, while the balanced condition enforces equal class sizes of 200 samples each. At N=1,000, a model that defaults to the majority class in the imbalanced setting would achieve 51.1 % accuracy with zero language understanding — motivating the need to report both conditions side by side.

3.3 Preprocessing Pipeline

The preprocessing steps, implemented in Python using pandas, are applied in the following order to every record:

1. Drop rows with empty `product` or `narrative`.
2. Retain only records belonging to the five target categories.
3. Remove narratives longer than 500 characters, reducing processing time and GPU memory pressure while avoiding truncation artifacts during inference.
4. Standardise narrative text: convert to lowercase, remove special characters, and normalise whitespace.

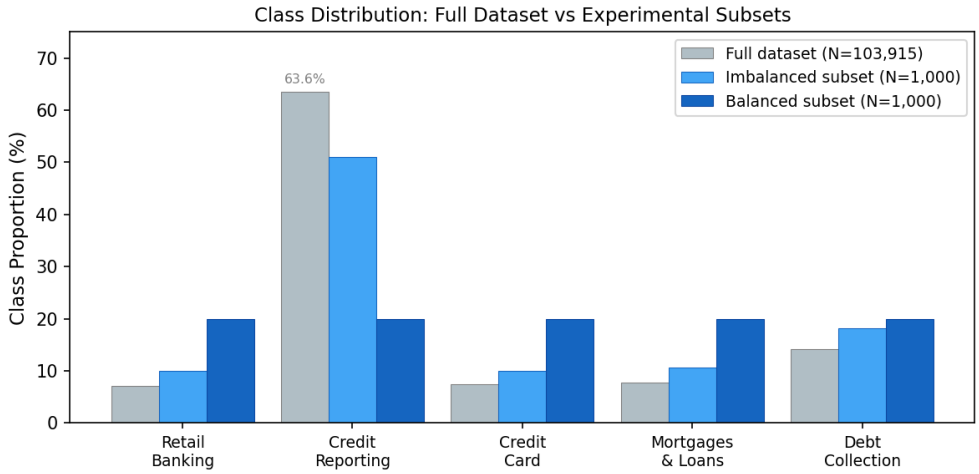


Fig. 1 Class proportions (%) in the full post-filter dataset (N=103,915), the imbalanced experimental subset (N=1,000), and the balanced subset (N=1,000). In the imbalanced condition, `credit_reporting` is still the dominant class at 51.1 %, confirming that balanced evaluation is essential to avoid majority-class inflation.

3.4 Experimental Subsets

Two experimental conditions are defined, each evaluated under a multi-seed, multi-size design to ensure statistically reliable estimates.

Experiment A (Balanced): 200 records are sampled uniformly at random from each of the five categories (N=1,000 total) using stratified undersampling. This follows the evaluation protocol of Roumeliotis et al. [10]; because the original sample identifiers are not publicly released, the comparison should be interpreted as protocol-matched rather than instance-identical.

Experiment B (Imbalanced): Records are sampled with proportions reflecting the post-filter natural class distribution, subject to a minimum of 100 samples per class. This yields N=1,000 with `credit_reporting` 51.1 %, `debt_collection` 18.2 %, `mortgages_and_loans` 10.7 %, `credit_card` 10.0 %, and `retail_banking` 10.0 %. The dominant `credit_reporting` class means a naive majority baseline would reach only 51.1 % accuracy with no language understanding. This second experiment, absent from the closest antecedent benchmark, assesses how class imbalance affects both accuracy estimates and per-class performance patterns.

Multi-seed and narrative-length design: Each condition is replicated with three independent random seeds (1, 2, and 3) and at two narrative character-length limits (250 and 500 characters), yielding 12 datasets per model and 132 inference runs in total (11 models $\times$ 3 seeds $\times$ 2 conditions $\times$ 2 length limits). All reported aggregate metrics and comparisons use the 500-character variant and are presented as mean $\pm$ standard deviation over the three seeds. The 250-character runs provide a narrative-length sensitivity check discussed in Section 5.5. This design addresses the single-seed limitation common in prior zero-shot LLM benchmarks and allows us to distinguish robust ranking patterns from seed-specific artefacts.

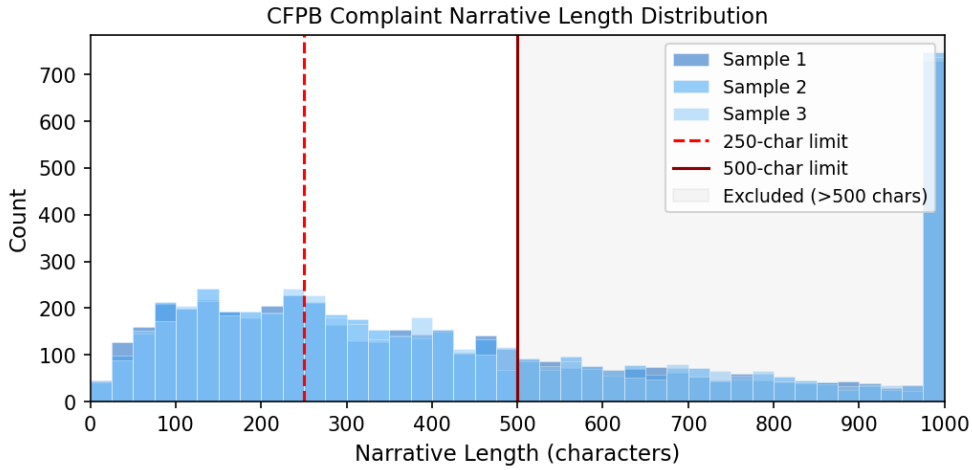


Fig. 2 Narrative character-length distribution across three balanced N=1,000 subsets (seeds 1–3, different colours). The dashed line marks the 250-character evaluation limit; the solid line marks 500 characters. Distributions are consistent across seeds, confirming reproducible sampling. Most narratives fall below 300 characters, justifying 250 as a meaningful short-text evaluation point and 500 as the full-length cutoff.

The choice of 250- and 500-character limits is motivated by the narrative length distribution of the corpus. Fig. 2 shows that the vast majority of usable narratives fall within the 0–500 character range across all three seeds, with the median near 250 characters. The two limits thus span the core of the distribution: the 250-character variant tests performance on short narratives (median length), while the 500-character variant includes the full range of narratives up to the preprocessing cutoff.

4 Experimental Setup

4.1 Infrastructure

All inference experiments are conducted on a local Ubuntu 24.04 workstation equipped with an Intel Core i7-14700 processor, 32 GB RAM, and an NVIDIA GeForce RTX 5060 Ti GPU with 16 GB VRAM. The software environment uses Python 3.12, Ollama for local LLM serving, and TensorFlow 2.18 for supporting machine-learning utilities. All models are pre-downloaded before inference, ensuring fully local execution without third-party API calls or transfer of complaint narratives to external services.

Model inference is handled by Ollama, a self-contained serving runtime that manages model loading, KV-cache management, and the local REST API. The Python inference script communicates with Ollama through the local endpoint, while evaluation scripts run in the same workstation environment. This configuration mirrors privacy-preserving deployments in which financial text remains on institution-controlled hardware.

Table 1 Open-Source Models Evaluated (Q4_K_M Quantisation Used in This Study)

Model	Family	Type	Params	VRAM (q4)
`gemma2:9b`	Google	Standard	9B	5.4 GB
`gemma4:12b`	Google	Standard	12B	7.6 GB
`mistral:7b`	Mistral	Standard	7B	4.4 GB
`mistral-nemo:12b`	Mistral	Standard	12B	7.1 GB
`llama3.1:8b`	Meta	Standard	8B	4.9 GB
`phi4:14b`	Microsoft	Standard	14B	9.1 GB
`deepseek-r1:8b`	DeepSeek	Reasoning	8B	5.2 GB
`deepseek-r1:14b`	DeepSeek	Reasoning	14B	9.0 GB
`qwen3:8b`	Alibaba	Reasoning	8B	5.2 GB
`qwen3:14b`	Alibaba	Reasoning	14B	9.3 GB
`gpt-oss:20b`	OpenAI	Reasoning	20B	13.0 GB

4.2 Models

Table 1 lists all eleven models. Six are standard (non-reasoning) instruction-tuned models; five are reasoning models that generate chain-of-thought traces before producing a final answer. The model set is designed to span two scale tiers within each family (approximately 7–9B and 12–14B parameters), enabling a parameter-matched cross-architecture comparison. The evaluated Ollama builds use 4-bit quantisation (Q4_K_M GGUF format, Ollama default), which reduces VRAM requirements significantly while maintaining competitive quality. The largest model, `gpt-oss:20b`, requires 13.0 GB VRAM and fits within the 16 GB local GPU memory budget. The evaluated model families build on recent open-weight model lines, including LLaMA [28], Mistral [29], Gemma [30], Phi [31], Qwen [24], and DeepSeek [22, 23]. Model names in Table 1 are the exact Ollama tags used in the experiments. The family column records the upstream lineage or registry metadata associated with each tag and should not be interpreted as equivalence to a proprietary API model with a similar provider name. All artifacts were obtained through `ollama pull <tag>` with no task-specific fine-tuning or local weight modification. `gpt-oss:20b` is an open-weight reasoning model released by OpenAI and distributed through the Ollama registry; it requires no API key and its weights are publicly accessible via `ollama pull gpt-oss:20b`. The "OpenAI" entry in the Family column reflects the model's originating developer, not access through any proprietary API or commercial service.

4.3 Prompt Design

A single, model-agnostic prompt is used for all models and all experiments. The prompt follows the five-label JSON-output template used by Roumeliotis et al. [10] to ensure comparability:

> *You are an AI assistant specializing in consumer complaint classification. Your task is to analyze a consumer complaint and classify it into the most appropriate category from the predefined list: ["retail_banking", "credit_reporting", "credit_card", "mortgages_and_loans",*

"debt_ collection"]. Provide your final classification in the following JSON format without explanations: `{"product": "chosen_category_name"}`. *Complaint: [NARRATIVE]*

The temperature parameter is set to 0 for all models to reduce sampling variability and improve reproducibility. No explicit `top_p`, `max_tokens`, stop-sequence, or model-specific thinking-mode flags are set; Ollama defaults are used for all other decoding parameters. Although the prompt asks for JSON without explanations, some reasoning-oriented builds may still emit intermediate text before the final answer. The inference script therefore extracts the final JSON object from the response by searching for the last valid `{"product": "..."}` occurrence using a regular expression, accounting for models that prepend reasoning traces before the answer.

The inference script was implemented using only Python standard-library modules (`csv`, `json`, `urllib`, `time`, `argparse`), eliminating the need for any `pip`-installed dependencies on the compute node. Each request is retried once on HTTP failure before being recorded as an error; errors are included in the final count and do not silently inflate accuracy. The raw model response is retained alongside the parsed label in the output CSV to allow post-hoc auditing of all parsing decisions. The full experiment was orchestrated as local batch runs totalling 132 tasks (11 models $\times$ 12 datasets), with runs scheduled sequentially or in controlled parallel groups according to GPU memory availability.

4.4 Evaluation Metrics

Following the closest antecedent benchmark, we compute four primary metrics over all N=1,000 predictions per run: *exact-match accuracy*, *weighted precision*, *weighted recall*, and *weighted F1-score*. We additionally report *macro-F1*, which weights all classes equally and is especially informative for the imbalanced experiment. Weighted variants weight each class by its sample size, making them appropriate for comparison with the reference paper but potentially optimistic under class imbalance. We additionally record the *mean latency* (seconds per prediction, wall-clock time) and the number of *off-label* predictions — responses where the model returned a category name not in the prescribed five, or failed to return valid JSON entirely. Off-label predictions are counted as errors. The API cost for all experiments is $0. Compute time is supplied by the local workstation rather than a commercial LLM API.

5 Results

5.1 Experiment A: Balanced Evaluation

Table 2 presents the full results for Experiment A, reported as means over three independent seeds. The `gemma4:12b` standard model achieves the highest accuracy (0.774 ± 0.017) and weighted F1 (0.774 ± 0.016), surpassing all reasoning models including the larger `gpt-oss:20b` (F1 0.749 ± 0.019). This is the most striking finding of the study: a 12B non-reasoning model outperforms a 20B reasoning model and ranks first among all eleven models tested. Second place goes to `qwen3:14b` (F1 0.750 ± 0.011), a reasoning model, followed by `gpt-oss:20b` in third. `deepseek-r1:14b` (F1 0.724 ± 0.007) and `qwen3:8b` (F1 0.717 ± 0.002) complete the top five. The standard `phi4:14b`

Table 2 Experiment A Results — Balanced (N = 1,000, mean of 3 seeds; std in parentheses)

Model	T	Acc.	F1w	F1m	Lat.(s)	OL
gemma4:12b	S	0.774(±.017)	0.774(±.016)	0.774	17.11	3.7
qwen3:14b	R	0.751(±.011)	0.750(±.011)	0.750	5.73	0.0
gpt-oss:20b	R	0.748(±.020)	0.749(±.019)	0.749	4.13	0.0
deepseek-r1:14b	R	0.724(±.006)	0.724(±.007)	0.724	9.33	1.3
qwen3:8b	R	0.717(±.003)	0.717(±.002)	0.717	5.19	0.0
phi4:14b	S	0.705(±.012)	0.706(±.011)	0.706	0.54	1.0
deepseek-r1:8b	R	0.684(±.013)	0.683(±.012)	0.683	5.14	2.0
gemma2:9b	S	0.680(±.006)	0.676(±.009)	0.676	0.67	0.3
mistral:7b	S	0.593(±.002)	0.587(±.004)	0.587	0.23	58.7
llama3.1:8b	S	0.578(±.005)	0.587(±.003)	0.587	2.97	66.7
mistral-nemo:12b	S	0.559(±.019)	0.540(±.025)	0.540	0.59	0.7

T: R=Reasoning, S=Standard. OL=Off-label mean. Lat.=Mean latency.

(F1 0.706 ± 0.011) ranks sixth. Notably, `deepseek-r1:8b` (rank 7, F1 0.683 ± 0.012) only narrowly exceeds the standard `gemma2:9b` (rank 8, F1 0.676 ± 0.009), confirming that 8B reasoning models do not automatically dominate similarly sized standard models. The weakest model is `mistral-nemo:12b` (F1 0.540 ± 0.025, rank 11), which despite being a 12B model scores below both `mistral:7b` (F1 0.587) and `llama3.1:8b` (F1 0.587), suggesting that the Mistral Nemo architecture is not well-suited to strict JSON output tasks in the zero-shot setting. The low standard deviations across most models — notably `qwen3:8b` (±0.002) and `mistral:7b` (±0.004) — confirm that the ranking is stable across independent samples.

The `OL` column in Table 2 is particularly informative. The three zero-off-label models among the top-five (`qwen3:14b`, `gpt-oss:20b`, and `qwen3:8b`) are all reasoning models. The two models with severe format-following failures are standard models: `llama3.1:8b` produces 67 off-label predictions and `mistral:7b` produces 59. Importantly, `gemma4:12b` (rank 1, standard) produces only 3.7 off-label predictions on average — nearly as low as the best reasoning models — demonstrating that newer-generation standard models can achieve strong instruction adherence without chain-of-thought. At the same time, `phi4:14b` is notable among standard models: it reaches F1 0.706 with only one off-label prediction and a mean latency of 0.54 s, making it the strongest low-latency alternative.

We perform post-hoc exact McNemar tests (binomial, one-sided) on paired correctness outcomes for each of the three seeds independently, applying a Bonferroni-adjusted threshold of α=0.0125 (four comparisons anchored on `gemma4:12b`). The advantage of `gemma4:12b` over `deepseek-r1:14b` and `qwen3:8b` is significant in all three seeds (p<0.001 each), providing robust statistical evidence for those rank separations. Comparisons with `qwen3:14b` and `gpt-oss:20b` reach significance in only one seed each (F1 gap ≤ 0.024), confirming that these three models form a statistical performance cluster without a consistent ordering at N=1,000 per seed. The two lowest-performing

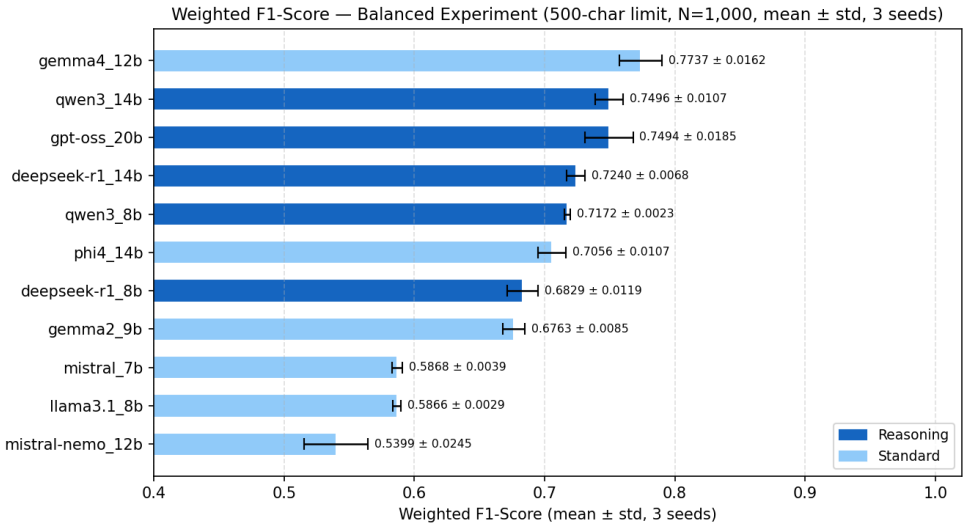


Fig. 3 Weighted F1-scores for all eleven models in Experiment A (Balanced), shown as mean ± std over three seeds. Reasoning models are shown in dark blue; standard models in light blue. `gemma4:12b` (standard) leads all reasoning models.

models (`llama3.1:8b`, `mistral-nemo:12b`) are clearly separated from all others by a wide margin.

For Experiment A, macro-F1 is numerically identical to weighted F1 because all five classes have equal support. Per-class inspection reveals that even the best model is uneven across labels: `gemma4:12b` reaches F1 0.825 on `retail_banking` and 0.820 on `mortgages_and_loans` but drops to 0.718 on `debt_collection` and 0.743 on `credit_reporting`. This motivates the macro-F1 reporting in Experiment B, where minority-class behaviour can otherwise be hidden by the majority class. Detailed per-class F1 for the leading models are reported in Table 5.

Fig. 3 visualises the F1-score distribution across models. Reasoning models (dark blue) cluster at the top while standard models show greater spread. The confusion matrix for the third-ranked and top reasoning model at 20B (`gpt-oss:20b`) is shown in Fig. 4. The model handles `credit_reporting` and `retail_banking` well but shows some confusion between `mortgages_and_loans` and `credit_card`. Fig. 5 shows the confusion matrix for `phi4:14b`, the best sub-1-second standard model: its strong diagonal confirms reliable five-class classification with only one off-label prediction, and minor confusion between `mortgages_and_loans` and `credit_card`. Fig. 6 shows the confusion matrix for `deepseek-r1:14b` (fourth overall, second among reasoning models): its near-zero off-label rate confirms that the reasoning trace enforces strict output format, and main confusions occur between `credit_card` and `mortgages_and_loans`. In contrast, `llama3.1:8b` (rank 10) produces 67 off-label predictions on average per seed — a large fraction of its samples are assigned invented categories, which directly degrades accuracy and recall across all five classes.

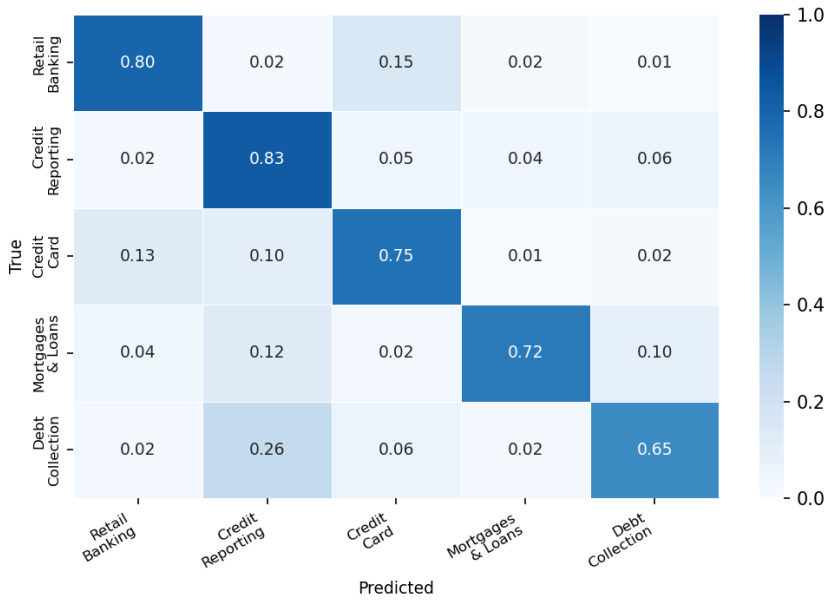


Fig. 4 Averaged confusion matrix for `gpt-oss:20b` (third-ranked model, top reasoning model, Experiment A, N=1,000, averaged over 3 seeds). Rows are true labels; columns are predicted labels.

5.2 Experiment B: Imbalanced Evaluation

Table 3 presents results for Experiment B, again as means over three seeds. All eleven models show numerically higher accuracy and F1 compared to Experiment A. `gpt-oss:20b` regains the top position under imbalanced conditions, with mean accuracy 0.782 ± 0.017 and F1 0.783 ± 0.016. `gemma4:12b` follows closely in second place (Acc 0.776 ± 0.015, F1 0.779 ± 0.014), and `qwen3:14b` ranks third (F1 0.772 ± 0.013). The top five models (`gpt-oss:20b`, `gemma4:12b`, `qwen3:14b`, `qwen3:8b`, `phi4:14b`) all achieve weighted F1 ≥ 0.754. The imbalanced setting changes the absolute score level more than the identity of the leading models. However, the gains for `llama3.1:8b` and `mistral:7b` should not be interpreted as genuine robustness improvements, because both models still produce many off-label outputs and benefit from the dominant `credit_reporting` class.

The relative model ranking is broadly preserved compared to Experiment A, with one notable reordering: `gpt-oss:20b` reclaims the top position while `gemma4:12b` drops to second. The macro-F1 column makes the imbalance effect visible: `gpt-oss:20b` has weighted F1 0.783 but macro-F1 0.740, while `llama3.1:8b` has weighted F1 0.647 but macro-F1 0.564. The gap shows that weighted metrics substantially overstate minority-class performance in the imbalanced setting. The F1-score distribution mirrors Experiment A in its broad ordering (Fig. 7 right bars), with `gpt-oss:20b`, `gemma4:12b`, and `qwen3:14b` forming the leading group.

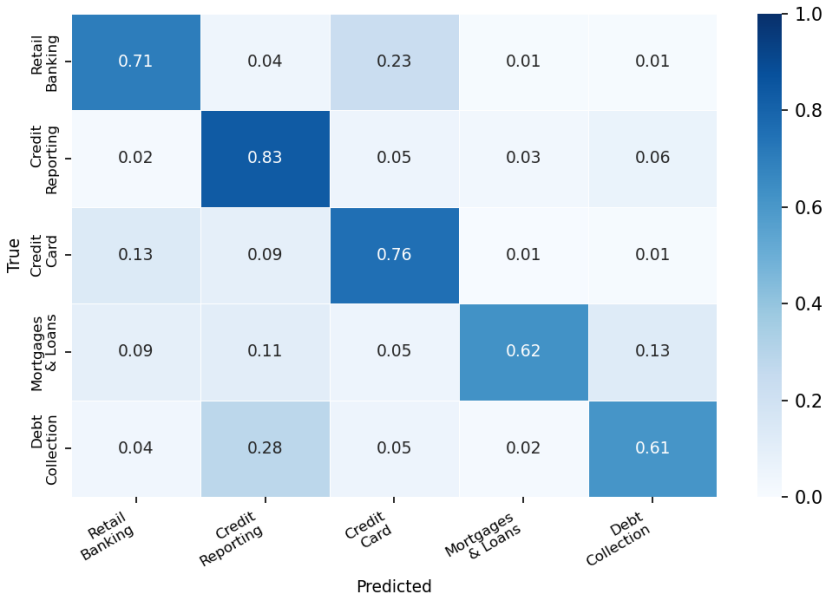


Fig. 5 Averaged confusion matrix for `phi4:14b` (sixth overall, best sub-1-second standard model, Experiment A, averaged over 3 seeds). Strong diagonal with minor confusion between `mortgages_and_loans` and `credit_card`.

Table 3 Experiment B Results — Imbalanced (N = 1,000, mean of 3 seeds; std in parentheses)

Model	T	Acc.	F1w	F1m	Lat.(s)	OL
`gpt-oss:20b`	R	0.782(±.017)	0.783(±.016)	0.740	4.01	0.0
`gemma4:12b`	S	0.776(±.015)	0.779(±.014)	0.741	10.61	2.7
`qwen3:14b`	R	0.772(±.012)	0.772(±.013)	0.730	5.53	0.0
`qwen3:8b`	R	0.766(±.008)	0.762(±.007)	0.719	6.97	0.3
`phi4:14b`	S	0.753(±.004)	0.754(±.002)	0.696	0.58	0.0
`deepseek-r1:14b`	R	0.746(±.017)	0.748(±.015)	0.700	7.32	0.3
`deepseek-r1:8b`	R	0.738(±.012)	0.731(±.011)	0.677	4.46	2.3
`gemma2:9b`	S	0.720(±.012)	0.719(±.014)	0.656	0.67	0.3
`mistral-nemo:12b`	S	0.653(±.011)	0.641(±.013)	0.538	0.59	0.7
`mistral:7b`	S	0.649(±.010)	0.661(±.010)	0.566	0.23	42.3
`llama3.1:8b`	S	0.632(±.006)	0.647(±.012)	0.564	0.56	57.3

T: R=Reasoning, S=Standard. OL=Off-label mean. Lat.=Mean latency.

5.3 Cross-Experiment Comparison

Fig. 7 provides a side-by-side view of accuracy across both experiments. All models improve numerically from Experiment A to Experiment B; the shift is most pronounced for `mistral:7b` (Δacc = +0.055) and `llama3.1:8b` (Δacc = +0.054), the two models with the highest off-label counts. This pattern suggests that under

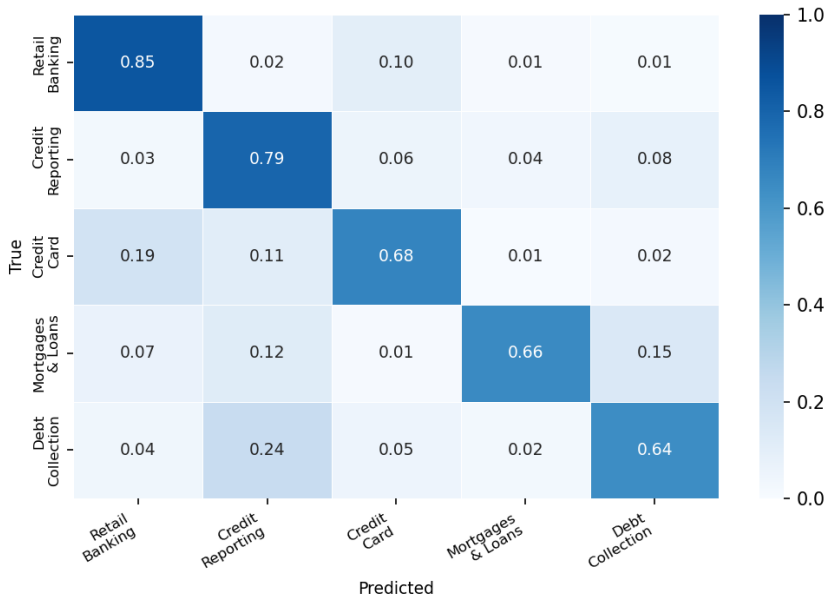


Fig. 6 Averaged confusion matrix for `deepseek-r1:14b` (fourth-ranked model, second among reasoning models, Experiment A, averaged over 3 seeds). Near-zero off-label rate; main confusions occur between `credit_card` and `mortgages_and_loans`.

imbalanced conditions, a model defaulting to the majority class can still achieve a deceptively high aggregate score. `mistral-nemo:12b` improves by +0.094 accuracy from Experiment A to B, the largest gain of all eleven models, further confirming that its Experiment A result reflects failure to maintain the balanced classification task rather than a model-quality ceiling.

5.4 Comparison with Paid-API Models

Fig. 8 compares all eleven of our open-source models with the five best reported results from Roumeliotis et al. [10]. Table 4 summarises the key comparison points. Our best model, `gemma4:12b`, achieves a mean accuracy of 0.774 ± 0.017 and F1 of 0.774 ± 0.016 — virtually indistinguishable from the paid GPT-4.5 (F1 0.7763) and within 0.003 of o1 (F1 0.7764) under the matched protocol, at zero API cost. This is the headline finding: a freely available, locally-deployed 12B standard model matches the performance of the two most capable paid reasoning models in the literature. `qwen3:14b` (F1 0.750) and `gpt-oss:20b` (F1 0.749) both exceed the paid GPT-4o baseline (0.7358), although they remain below Claude 3.7 Sonnet (F1 0.7626). Because the reference paper does not publish the exact sampled complaint identifiers, this comparison should be read as protocol-matched rather than paired on identical instances. The cost differential is stark: the reference paper reported o1 costs of $24.24 per thousand predictions, versus $0.00 for all our models. Thus, Table 4 supports a strong result: the best local model (`gemma4:12b`) matches frontier paid reasoning models on this task while eliminating all API cost and data transfer to third parties.

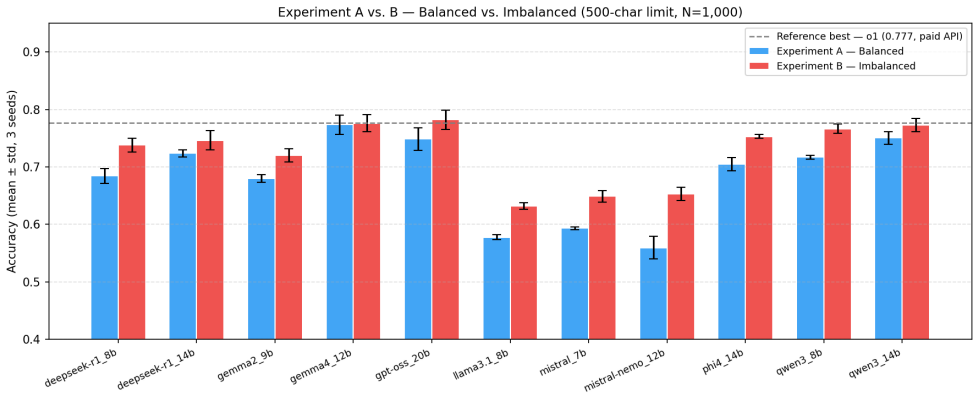


Fig. 7 Side-by-side comparison of accuracy (mean ± std, 3 seeds) across Experiment A (Balanced, blue) and Experiment B (Imbalanced, red) for all models (N=1,000). The dashed reference line marks the best published paid-API result (o1, 0.777). The upward shift in Experiment B partially reflects class imbalance rather than genuine model improvement.

Table 4 Zero-Shot Performance vs. API Cost (Balanced, N=1,000). Commercial API figures are from [10].

Model	Source	Accuracy	F1	Cost/1k
o1	Paid API	0.777	0.7764	$24.24
GPT-4.5	Paid API	0.774	0.7763	$12.32
`gemma4:12b`	**Open-source**	**0.774±.017**	**0.774±.016**	**$0.00**
Claude 3.7 Sonnet	Paid API	0.765	0.7626	$0.64
Gemini 2.0 Flash	Paid API	0.758	0.7586	$0.04
`qwen3:14b`	Open-source	0.751±.011	0.750±.011	$0.00
`gpt-oss:20b`	Open-source	0.748±.020	0.749±.019	$0.00
GPT-4o	Paid API	0.736	0.7358	$0.45
`deepseek-r1:14b`	Open-source	0.724±.006	0.724±.007	$0.00
`qwen3:8b`	Open-source	0.717±.003	0.717±.002	$0.00
`phi4:14b`	Open-source	0.705±.012	0.706±.011	$0.00

5.5 Narrative Length Sensitivity

Fig. 9 compares mean accuracy on 250-character-limit versus 500-character-limit narratives for the balanced condition (both at N=1,000). The overall ranking is highly stable: no model changes its relative rank by more than one position between the two length limits. Most models improve by 0–3 percentage points moving from shorter to longer narratives, with `deepseek-r1:14b` (+1.6 pp) and `llama3.1:8b` (+2.8 pp) showing the largest gains, suggesting these models benefit from the additional context available in longer narratives. `gpt-oss:20b` and `mistral-nemo:12b` show virtually no change (≤0.1 pp), indicating that their behaviour is not sensitive to narrative length

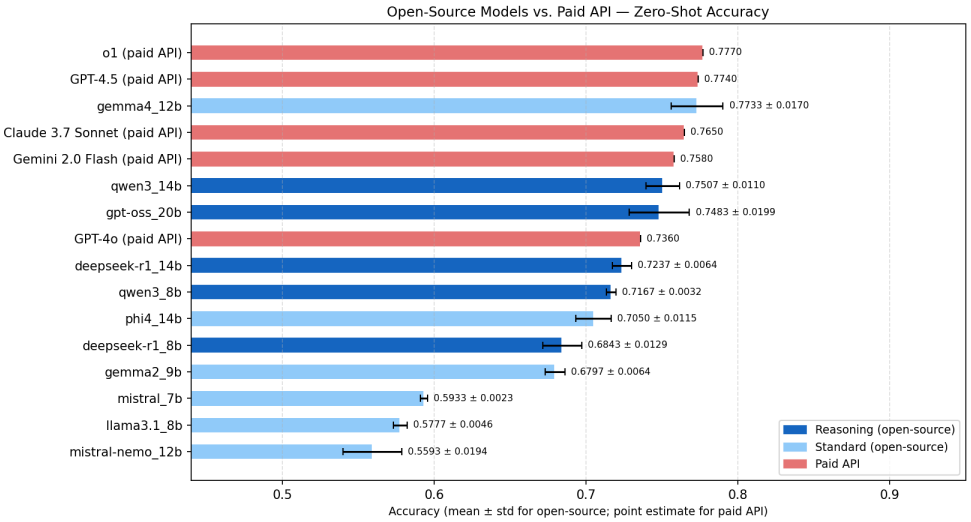


Fig. 8 Accuracy of all eleven open-source models (Experiment A, mean ± std over 3 seeds) alongside five paid-API baselines from the literature [10]. `gemma4:12b` matches GPT-4.5 and approaches o1 at zero cost. Open-source error bars reflect seed variability; paid-API figures are single-run point estimates.

in this range. The stability of the top-cluster ranking confirms that the 500-character results represent genuine model-quality differences and not length-specific artefacts.

5.6 Latency Analysis

Mean per-prediction latency is shown in the Lat.(s) column of Tables 2 and 3. Standard models are notably faster: `mistral:7b` completes a prediction in 0.23 s on average, while `phi4:14b` requires only 0.54 s despite being twice the parameter count. Reasoning models are slower: `deepseek-r1:14b` takes 9.33 s per prediction due to its chain-of-thought trace. `gemma4:12b`, despite being a standard model, has the highest latency at 17.1 s — attributable to its longer response patterns rather than chain-of-thought. The latency–accuracy trade-off is model-type dependent: `phi4:14b` achieves the best latency-accuracy ratio among all models (0.54 s, F1 0.706). Fig. 10 shows the F1-score distribution across all eleven models in Experiment B, providing a visual complement to Table 3.

5.7 Per-Class Breakdown for Leading Models

Table 5 reports per-class F1 for the three leading models in each experiment, confirming that aggregate weighted F1 masks important minority-class variation. In Experiment B all three models jump to $\geq$0.84 F1 on `credit_reporting` while minority classes remain in the 0.65–0.77 range, confirming the majority-class bias effect discussed in Section 6.

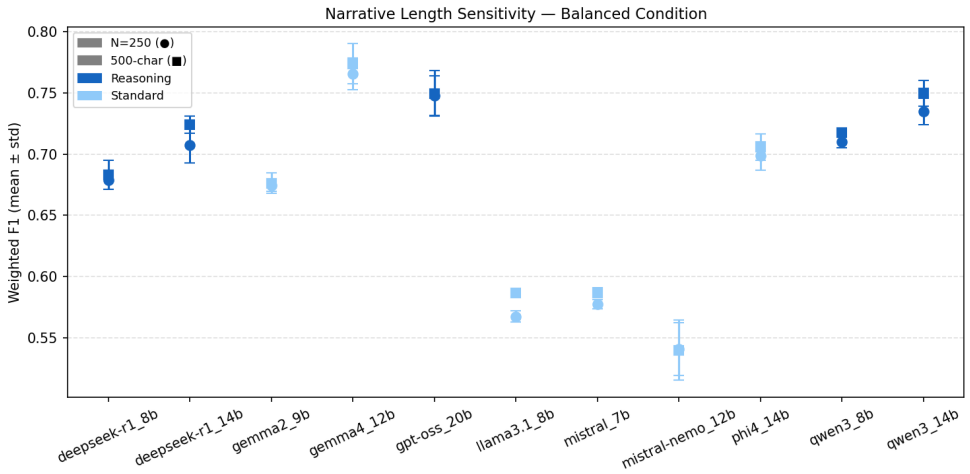


Fig. 9 Mean accuracy on 250-char vs 500-char narrative variants (Balanced, N=1,000, 3 seeds) for all eleven models. Rankings are highly stable; most models show small gains with longer narratives.

6 Discussion

6.1 Reasoning vs. Standard Models

The most notable finding across both experiments is that model *generation* and *architecture quality* can outweigh model type (reasoning vs. standard). `gemma4:12b`,

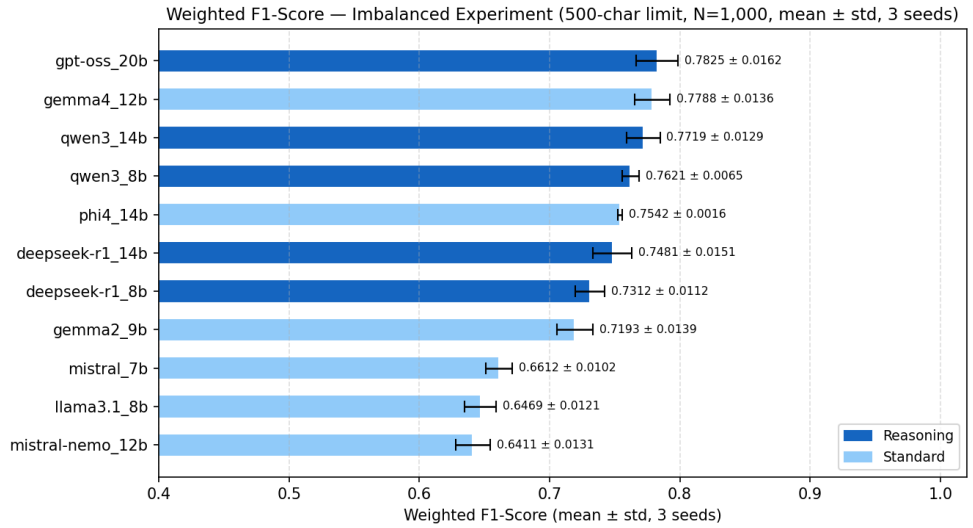


Fig. 10 Weighted F1-scores for all eleven models in Experiment B (Imbalanced), shown as mean ± std over three seeds (N=1,000). Reasoning models are shown in dark blue; standard models in light blue. `gpt-oss:20b` leads, with `gemma4:12b` a close second.

Table 5 Per-Class F1 for Leading Models (Exp. A and B, N=1,000)

E	Model	Retail	CreditRep.	Card	Mortg.	Debt
A	`gemma4:12b`	0.8247	0.7432	0.7634	0.8194	0.7178
A	`qwen3:14b`	0.7855	0.7212	0.7296	0.8097	0.7019
A	`gpt-oss:20b`	0.7939	0.7137	0.7394	0.7953	0.7045
B	`gpt-oss:20b`	0.7508	0.8527	0.6657	0.7404	0.6919
B	`gemma4:12b`	0.7676	0.8453	0.6719	0.7442	0.6769
B	`qwen3:14b`	0.7353	0.8437	0.6544	0.7427	0.6727

E = Experiment; Mortg. = Mortgages & Loans.

a standard non-reasoning model, achieves the highest weighted F1 in Experiment A (0.774 ± 0.016), surpassing all five reasoning models including `gpt-oss:20b` (F1 0.749), which has 67 % more parameters. This challenges the simple hypothesis that reasoning models are necessary for structured zero-shot classification tasks.

That said, a nuanced pattern emerges when scale is controlled. At the 8B tier, reasoning models (`deepseek-r1:8b`, `qwen3:8b`) generally outperform standard models of similar size (`gemma2:9b`, `llama3.1:8b`, `mistral:7b`). At the 14B tier, `qwen3:14b` (Reasoning, F1 0.750) and `phi4:14b` (Standard, F1 0.706) are both strong, but the reasoning advantage is narrower. The 12B result from `gemma4:12b` suggests that the latest-generation standard models have closed the reasoning gap through improved alignment and instruction-following training, consistent with the broader evidence that chain-of-thought benefits diminish as standard-model quality improves [21, 23].

An unexpected finding is the underperformance of `mistral-nemo:12b` (F1 0.540 ± 0.025 in Experiment A, rank 11 of 11), which falls below both its smaller sibling `mistral:7b` (F1 0.587) and all other 12B-tier models. Its off-label rate is low (0.7 per run), indicating that the model does produce parseable JSON but assigns incorrect categories. This suggests that the Mistral Nemo architecture, while competitive in general-purpose tasks, does not transfer as effectively to this narrow structured classification setting.

6.2 Within-Family Performance Patterns

The expanded eleven-model set allows direct within-family comparisons that were not possible in the original eight-model design.

Gemma family (Google): `gemma4:12b` (F1 0.774) outperforms `gemma2:9b` (F1 0.676) by 0.098 F1 points — the largest within-family gain observed. This improvement is driven by stronger category disambiguation. Off-label counts increase slightly from gemma2 (mean 0.3) to gemma4 (mean 3.7), but gemma4's rare off-label cases are classification mismatches rather than format failures — a qualitatively different error mode that is less disruptive to downstream parsing. The generation gap (Gemma 2 to Gemma 4) is more impactful than the parameter gap (9B to 12B), confirming that training recipe and alignment improvements can outweigh raw scale.

Mistral family (Mistral AI): `mistral-nemo:12b` (F1 0.540) performs *worse* than `mistral:7b` (F1 0.587) in Experiment A, despite having 71 % more parameters. Their error modes are opposite: `mistral:7b` frequently produces off-label outputs (58.7 per run), whereas `mistral-nemo:12b` commits to valid categories but assigns them incorrectly. Latency is comparable (0.23 s vs 0.59 s). This suggests that Mistral Nemo's training optimised for different task types and that the 12B Nemo architecture does not generalise as well as Mistral 7B to tightly constrained JSON classification.

Qwen3 family (Alibaba): `qwen3:14b` (F1 0.750) outperforms `qwen3:8b` (F1 0.717) consistently across both experiments, with no off-label predictions in either case. The +0.033 F1 gain is achieved with comparable latency (5.73 s vs 5.19 s), making `qwen3:14b` the better choice when additional VRAM is available (9.3 GB vs 5.2 GB). Both models demonstrate that the Qwen3 architecture transfers well to structured zero-shot tasks.

DeepSeek-R1 family (DeepSeek): `deepseek-r1:14b` (F1 0.724) outperforms `deepseek-r1:8b` (F1 0.683) by 0.041 F1 points. Off-label counts also improve with scale (1.3 vs 2.0). Latency increases proportionally (9.33 s vs 5.14 s), reflecting the longer reasoning traces generated by the larger model. The performance gap is consistent across all three seeds, confirming that scale reliably improves the DeepSeek-R1 distillation quality.

6.3 Off-Label Prediction Behaviour

Off-label predictions — responses where the model returned a label not in the prescribed five categories — are a direct measure of instruction-following reliability. The two models with the highest off-label counts in Experiment A (`llama3.1:8b`: 67; `mistral:7b`: 59) also rank among the lowest in F1. This is not coincidental: off-label responses are counted as prediction errors, directly reducing accuracy and recall. In Experiment A, the reasoning models are generally more disciplined than the weakest standard models: `qwen3:14b`, `gpt-oss:20b`, and `qwen3:8b` produce zero off-label predictions, while `deepseek-r1:14b` and `deepseek-r1:8b` remain low at 1.3 and 2.0 off-label predictions per run, respectively. The better instruction-following of reasoning models reflects the structured nature of chain-of-thought: the model explicitly reasons about which label is appropriate before formatting its output, reducing the likelihood of hallucinating a novel category.

The raw status logs show that off-label behaviour has two distinct forms. `llama3.1:8b` primarily fails at the formatting/parsing level, producing a mean of 67 parse errors in Experiment A and 57 in Experiment B. In contrast, `mistral:7b` usually returns parseable JSON but invents labels outside the allowed taxonomy, producing a mean of 59 valid-but-out-of-vocabulary labels in Experiment A and 42 in Experiment B. Smaller counts for `phi4:14b`, `gemma2:9b`, and `deepseek-r1:8b` are also valid-but-out-of-vocabulary labels such as `unknown` or `health_insurance`. This distinction matters operationally: parse errors indicate output-format failure, whereas out-of-vocabulary labels indicate failure to stay within the regulatory taxonomy.

Off-label behaviour should therefore be interpreted not only as an accuracy penalty but also as an operational reliability signal. In a regulated complaint-routing pipeline, an invented category such as `unknown`, `loan`, or `credit_dispute` may be more disruptive

than an ordinary within-taxonomy misclassification, because downstream systems typically assume that every prediction belongs to the fixed regulatory label set. The observed difference between parse errors and valid-but-out-of-taxonomy labels suggests that local LLM evaluations should report output discipline separately from classification quality, especially when models are intended for semi-automated banking, compliance, or regulator-facing workflows.

Notably, the off-label count for `mistral:7b` improves substantially in Experiment B (from 59 to 42), suggesting that the dominant `credit_reporting` class provides an easier default that is more likely to receive a correctly-formatted taxonomy label. By contrast, `llama3.1:8b` remains unreliable (67 in Experiment A vs 57 in Experiment B), indicating that its parsing failures are not resolved by class imbalance alone.

6.4 Latency–Accuracy Trade-off

From a practical deployment perspective, the latency results present an important consideration. The fastest model, `mistral:7b` (0.23 s), achieves only F1 0.587 — far below the leading models in Experiment A. `phi4:14b`, at 0.54 s per prediction and F1 0.706, offers the most favourable accuracy-to-latency ratio of any model tested. For applications requiring both real-time response and reasonable accuracy, `phi4:14b` is the recommended choice.

Reasoning models are consistently slower: `deepseek-r1:14b` takes 9.33 s per prediction on average across seeds on the local workstation. At 500 predictions, this translates to roughly 1.3 hours of total inference time on the RTX 5060 Ti. For large-scale regulatory applications processing millions of complaints, the time budget would need to be considered alongside model quality. All reported latency figures for reasoning models include the full chain-of-thought generation trace; the final JSON label is extracted from the end of this trace via the regex described in Section 4. Latency comparisons between reasoning and standard models should therefore be interpreted as end-to-end wall-clock cost rather than pure decoding speed.

6.5 Deployment Recommendations

Based on the combined latency and accuracy evidence, we offer the following concrete deployment recommendations for practitioners.

Maximum accuracy (offline / batch): Use `gemma4:12b`. It achieves the highest balanced F1 (0.774), matches paid frontier APIs at zero cost, and requires 7.6 GB VRAM. Latency of 17.1 s per prediction is acceptable for batch triage of large complaint archives.

Real-time deployment (<1 s latency required): Use `phi4:14b`. It offers F1 0.706 at just 0.54 s per prediction — the best accuracy-to-latency ratio among all eleven models. At roughly 110 predictions per minute throughput, it can serve a moderate complaint routing queue in real time on the local GPU.

Constrained VRAM (≤5.5 GB): Use `qwen3:8b` (5.2 GB, F1 0.717, 5.19 s). It produces zero off-label predictions and outperforms all other sub-6 GB models.

Avoid `mistral:7b` and `llama3.1:8b` in zero-shot settings: both produce >50 off-label predictions per 1,000 samples on average, making them operationally unreliable

without post-processing. `mistral-nemo:12b` is also not recommended for this task despite its size advantage.

Taken together, these recommendations translate into a concrete cost advantage. At moderate scale (one million complaints per year), deploying `gemma4:12b` instead of the best-performing paid API model would avoid roughly $12,320 in API fees with no measurable quality loss. `qwen3:14b` and `gpt-oss:20b` similarly exceed the paid GPT-4o baseline (F1 0.7358) at zero API cost. This should not be read as computation being free: local inference still consumes GPU time, electricity, and local hardware resources. The practical gain is the elimination of recurring third-party API charges and the need to transfer sensitive complaint text to external services.

6.6 Interpreting the Imbalanced Results

The across-the-board increase in accuracy and F1 from Experiment A to Experiment B demands careful interpretation. While a genuine improvement is possible — models may be aided by the greater frequency of `credit_reporting` training signal in their pre-training corpora, which coincides with its frequency in the CFPB data — a more parsimonious explanation exists: *class imbalance inflates aggregate accuracy metrics.*

When 51.1 % of the test set belongs to a single class, a model that classifies most ambiguous narratives as `credit_reporting` will achieve a high weighted accuracy even if its per-class performance is poor. The particularly pronounced gains for `llama3.1:8b` (+4.2 pp accuracy) and `mistral:7b` (+5.5 pp accuracy) — the two highest off-label producers in Experiment A — are consistent with this mechanism: under imbalanced conditions, a confused model benefits disproportionately from majority-class guesses. We therefore caution against using Experiment B accuracy as a simple indicator of model quality and recommend that future evaluations report per-class precision and recall alongside aggregate metrics.

6.7 Limitations

This study has several limitations. First, the balanced comparison matches the reference protocol but not necessarily the exact sampled instances, since the original sample identifiers are unavailable. Second, the 500-character length filter reduces inference cost but also shifts the class distribution, especially increasing the share of `credit_reporting`; future work should test whether the same patterns hold for longer narratives. Third, lowercasing and special- character removal may discard useful cues such as account-number patterns, capitalization, dates, or punctuation; the effect of less aggressive LLM-oriented preprocessing remains untested. Fourth, all experiments use N=1,000 examples per seed, evaluated at two narrative character-length limits (250 and 500 characters) across a five-class taxonomy; larger sample sizes, longer narratives, and richer taxonomies may reveal additional drift and rarer category confusions. Fifth, all local models are evaluated with a single zero-shot prompt. Prompt wording, output-format instructions, few-shot examples, or self-consistency decoding may change the relative ranking of models, particularly for smaller standard models with weaker instruction following. Sixth, all local models are evaluated in their q4 Ollama quantised form, so the reported scores conflate model family, model size, and quantisation effects.

Seventh, we cannot rule out pre-training contamination: open-weight models may have encountered CFPB complaints or derivative datasets during pre-training, although no task-specific fine-tuning is used in our experiments. Eighth, the reported $0 cost refers to API expenditure, not to the full economic cost of GPU hardware, electricity, or local workstation operation. Finally, the main tables report weighted aggregate metrics; for operational deployment, macro-F1 and per-class precision/recall should be inspected to avoid masking poor minority-class performance.

7 Conclusion

We have presented a comprehensive zero-shot benchmark of eleven open-source large language models on the CFPB consumer financial complaint classification task, conducted through local self-hosted inference at zero API cost. Our results demonstrate that the best open-source model, `gemma4:12b`, achieves a mean weighted F1 of 0.774 ± 0.016 across three independent seeds on the balanced evaluation — matching the paid GPT-4.5 (F1 0.7763) and approaching o1 (F1 0.7764) at zero API cost. The multi-seed design confirms that this ranking is statistically stable: standard deviations across seeds are ≤ 0.02 for all models, and the top cluster (`gemma4:12b`, `qwen3:14b`, `gpt-oss:20b`) is consistent across all three seeds. Notably, the top-performing model is a standard (non-reasoning) 12B model, demonstrating that the latest generation of standard instruction-tuned models can match or surpass larger reasoning models on structured classification tasks. The standard model `phi4:14b` (F1 0.706 ± 0.011) offers the best latency–accuracy trade-off at 0.54 s per prediction.

Our secondary imbalanced experiment reveals important evaluation dynamics: aggregate accuracy improvements under imbalanced conditions are partially attributable to majority-class bias rather than genuine model improvement, underscoring the need for per-class reporting in real-world settings.

From a practical standpoint, these findings validate the viability of self-hosted LLM deployments for financial NLP tasks: institutions that cannot share sensitive complaint text with third-party API providers, or that operate under budget constraints, can now achieve competitive classification quality using open-weight models on standard GPU hardware.

Future work will explore five directions. First, a *quantisation sensitivity study* will evaluate all eleven models at q4, q8, and fp16 precision to determine whether the accuracy gains from higher numerical precision justify the increased VRAM cost — a sub-study not attempted in the reference paper. Second, a *prompt sensitivity study* will repeat the balanced and imbalanced evaluations across prompt variants, reporting confidence intervals and paired significance tests beyond the multi-seed McNemar analysis presented here. Third, and most directly motivated by the present results, we will *fine-tune the same eleven open-weight models* on the CFPB training split and compare their supervised performance against their zero-shot counterparts reported here. This paired comparison will quantify the exact cost of removing the annotation requirement and clarify whether the remaining small gap between local models and frontier paid APIs can be closed by task-specific supervised training. Fourth, *long-narrative robustness* will be examined by relaxing the 500-character filter

and measuring whether additional complaint context improves classification or instead increases prompt length and latency costs. Fifth, the benchmark will be *extended to the full multi-year CFPB corpus* with a richer category taxonomy to assess scalability and temporal robustness of the open-source inference pipeline.

Declarations

Funding. This research received no external funding.

Conflict of interest. The authors declare that they have no conflict of interest.

Ethics approval and consent to participate. Not applicable.

Consent for publication. Not applicable.

Data availability. The study uses the publicly available CFPB Consumer Complaints dataset published on Kaggle [27].

Materials availability. Not applicable.

Code availability. All source code, preprocessing scripts, and local experiment files used in this study are publicly available at: https://github.com/muhammettan28/ConsumerComplaint

Author contributions. Muhammet Tan and Halit Bakır contributed to the study conception, experimental design, implementation, analysis, and manuscript preparation.

References

[1] Das, S., *et al.*: Explainable AI for financial consumer complaint classification. IEEE Transactions on Computational Social Systems **11**, 3606–3615 (2024)

[2] Liu, Y., Ouyang, X., Bai, R., Zhu, X., Zhou, Y., Zhang, Y., Liu, C.: A framework for categorizing complaint text via large language model. In: Proceedings of the 7th International Conference on Advanced Algorithms and Control Engineering (ICAACE) (2024). IEEE

[3] Jondhale, R., Patil, S., Shinde, A., Ajalkar, D., Biradar, S.: Predicting consumer complaint disputes in finance using machine learning. In: Proceedings of the Second International Conference on Advances in Information Technology (ICAIT) (2024). IEEE

[4] Di, Z., Xiao, T., Li, B., Peng, Y., Li, J., Cheng, X., Xu, L., Zhu, X., Zheng, H.: Research on mobile network user complaint warning method based on multimodal data. In: 2023 International Conference on Information and Communication Technologies for Disaster Management (ICT-DM), pp. 1–6 (2023). https://doi.org/10.1109/ICT-DM58371.2023.10286947 . IEEE

[5] Devlin, J., Chang, M.-W., Lee, K., Toutanova, K.: BERT: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019

Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT), pp. 4171–4186 (2019). Association for Computational Linguistics

[6] Vairetti, C., *et al.*: BETO-based prioritization of consumer complaints. European Journal of Operational Research **312**, 1108–1118 (2024)

[7] Brown, T.B., *et al.*: Language models are few-shot learners. In: Advances in Neural Information Processing Systems (NeurIPS), vol. 33, pp. 1877–1901 (2020)

[8] Kojima, T., Gu, S.S., Reid, M., Matsuo, Y., Iwasawa, Y.: Large language models are zero-shot reasoners. Advances in Neural Information Processing Systems (NeurIPS) **35** (2022)

[9] Madanchian, M.: Generative AI for consumer behavior prediction: Techniques and applications. Sustainability **16** (2024)

[10] Roumeliotis, K.I., Tselikas, N.D., Nasiopoulos, D.K.: Think before you classify: The Rise of Reasoning Large Language Models for Consumer Complaint Detection and Classification. Electronics **14**(6), 1070 (2025) https://doi.org/10.3390/electronics14061070

[11] Raju, A., *et al.*: Topic modeling and classification of CFPB consumer complaints using transformer models. In: Proceedings of IEEE International Conference on Machine Learning and Applications (2022). IEEE

[12] Lilleberg, J., Zhu, Y., Zhang, Y.: Support vector machines and Word2vec for text classification with semantic features. In: 2015 IEEE 14th International Conference on Cognitive Informatics & Cognitive Computing (ICCI*CC), pp. 136–140 (2015). https://doi.org/10.1109/ICCI-CC.2015.7259377 . IEEE

[13] Grootendorst, M.: BERTopic: Leveraging BERT and c-TF-IDF to Create Easily Interpretable Topics. Zenodo (2020). https://doi.org/10.5281/zenodo.4381785

[14] Zhao, W.X., et al.: A survey of large language models. arXiv preprint arXiv:2303.18223 (2023)

[15] Correa, D., *et al.*: Generative AI for consumer complaint analysis: Classification, summarization, and response generation. In: Proceedings of the IEEE Andescon (2024). IEEE

[16] Roumeliotis, K.I., Tselikas, N.D., Nasiopoulos, D.K.: Leveraging large language models in tourism: A comparative study of the latest GPT omni models and BERT NLP for customer review classification and sentiment analysis. Information **15** (2024)

[17] Roumeliotis, K.I., Tselikas, N.D., Nasiopoulos, D.K.: LLMs in e-commerce: A

comparative analysis of GPT and LLaMA models in product review evaluation. Natural Language Processing Journal **6**, 100056 (2024) https://doi.org/10.1016/j.nlp.2024.100056

[18] Li, F., Wang, Y., Xu, X., Wang, S., Liang, J., Chen, Z., Liu, W., Feng, Q., Duan, T., Huang, Y., Song, Q., Li, X.: Performance evaluations of large language models for customer service. International Journal of Machine Learning and Cybernetics (2024)

[19] Shin, J., et al.: Detecting LLM-written consumer complaints at scale. arXiv preprint arXiv:2311.16466 (2025). arXiv:2311.16466v4

[20] Liang, P., et al.: LLM-assisted consumer complaint writing: Evidence from CFPB data. arXiv preprint arXiv:2502.09747 (2025)

[21] Wei, J., et al.: Chain-of-thought prompting elicits reasoning in large language models. Advances in Neural Information Processing Systems (NeurIPS) **35** (2022)

[22] DeepSeek-AI, Bi, X., Chen, D., Chen, G., Chen, S., Dai, D., Deng, C., Ding, H., Dong, K., et al.: DeepSeek LLM: Scaling open-source language models with longtermism. arXiv preprint arXiv:2401.02954 (2024)

[23] Guo, D., et al.: DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. arXiv preprint arXiv:2501.12948 (2025)

[24] Qwen Team: Qwen3 technical report. arXiv preprint arXiv:2505.09388 (2025)

[25] Bulgakov, A., Turchin, A.: Zero-shot reasoning for consumer complaint outcome prediction with DeepSeek-R1. arXiv preprint arXiv:2601.16406 (2026)

[26] Güllü, B., *et al.*: Fine-tuning LLaMA-3.1-8B for consumer complaint classification. In: Proceedings of the IEEE International Conference on Artificial Intelligence and Computer Science (ACDSA), Antalya, Turkey (2025). IEEE

[27] Tiwari, S.: Consumer Complaints Dataset for NLP. Kaggle. Accessed: 15 June 2026 (2021). https://www.kaggle.com/datasets/shashwatwork/consume-complaints-dataset-fo-nlp

[28] Touvron, H., et al.: Llama 2: Open foundation and fine-tuned chat models. arXiv preprint arXiv:2307.09288 (2023)

[29] Jiang, A.Q., et al.: Mistral 7B. arXiv preprint arXiv:2310.06825 (2023)

[30] Google DeepMind Gemma Team: Gemma 2: Improving open language models at a practical size. arXiv preprint arXiv:2408.00118 (2024)

[31] Abdin, M., et al.: Phi-4 technical report. arXiv preprint arXiv:2412.08905 (2024)